

Adaptive Multimodal Intelligence for Governance Crisis

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ABSTRACT —

Data silos, as well as multi-modal document formats and retrieval systems, pose serious obstacles to the management and utilisation of information by government entities. Current systems cannot fully understand semantics or perform cross-modality reasoning for complicated queries, leading to slow decision-making cycles or operational inefficiency. The following paper presents an intelligent system, the Unified Knowledge Platform for Government Centres, which leverages a state-of-the-art Multimodal Retrieval Augmented Generation (RAG) architecture. We extend beyond traditional text-based search by including audio, text, image, and video analysis within a unified knowledge graph. Key innovations include: Temporal and Event-Based Retrieval, through which all content is timestamped to provide a rich history of analysis; Multimodal Prompt Chaining, allowing users to develop connected narratives by reasoning across data types in a sequential manner; and an Adaptive Active Cycle forcing continuous improvement with the models given user input. It is a highly feasible system, modular in design, thus allowing offline functionality at the far end, and robust in risk management mechanisms, including encryption and fallback systems. This research creates a new paradigm for dynamic, intelligent, and trustworthy public sector knowledge management.

Keywords: Multimodal RAG, Government Knowledge Management, Cross-Modal Reasoning, Temporal Retrieval, Adaptive Learning.

1. INTRODUCTION

[1]Government work relies heavily on information, which emanates from various types of documents, audio files, video meetings, and scanned records. This information is key in decision-making, service provision, and transparency to the public.[4] However, much of this data resides within disparate locations and in different formats that do not work well together. Finding and utilising the information efficiently with manual efforts and simple search tools is quite difficult. These methods are slow, error-prone, and cannot perceive the

connectedness of diverse pieces of information, such as how a policy document, its related chart, and a meeting recording are related.

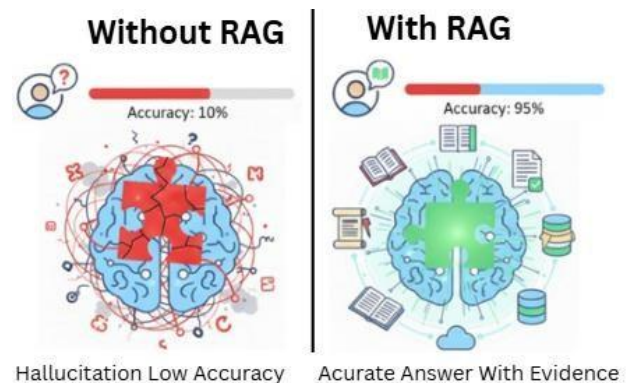


FIG.1.1 LLM ALONE VS RAG

It could be achieved using Artificial Intelligence, especially Large Language Models and Retrieval-Augmented Generation. RAG utilises real facts to make answers more accurate. Still, the standard RAG models work with text only and cannot process every kind of government data or understand when information has been created or shared. To remedy these shortcomings, we propose a Unified Knowledge Platform that transforms how information is handled within the government.

[8]While it is primarily a search tool, this particular platform is a comprehensive knowledge-gathering and utilisation system. It integrates sources of data-audio, images, videos, and documents-into a single, cohesive index. Therefore, people will be able to ask natural questions that are relevant within their context and receive helpful answers.

The major components of this system include:

1. Time- and Event-Based Searching:

Our system monitors times and events. By date or significant events, users can trace information. This helps to track policies and study past events.

2. Multimodal Prompt Chaining:

We utilise a special way of decomposing complex questions into sub-questions that can be answered by different kinds of data. This helps to elicit various sources of information together and create full and detailed answers.

3. Adaptive Learning Loop:

The system, over time, learns from user feedback. The learning process improves the finding and creation of information, thereby leading to accuracy and long-term utility. It focuses on being flexible, secure, and able to function without an internet connection. The latter is also important for its government use. Piecing it all together, this platform can make the work of governments faster, the decisions better supported by data, and more open and cooperative.

2. LITERATURE REVIEW

[3]Document management systems have evolved through several important stages. Early document management systems primarily relied on the use of Optical Character Recognition and keyword-based indexing. Such methods succeeded in digitising paper documents and thus enabling basic search functionality, but they lacked semantic understanding and consequently could not connect related information across different document types or formats. [10]One key step forward was the use of RAG systems, or Retrieval-Augmented Generation, which combined document retrieval with generative language models to provide more source-based answers and reduce the factual errors in generated responses. These initial RAG models were designed mostly for text data and failed to handle multimodal content. More recently, multimodal document AI systems are appearing that can process the textual and visual elements of documents.[15] These recognise that document analysis requires more than text processing per se, but the processing of layout, images, and other visual components. However, despite this, state-of-the-art systems currently clearly suffer from significant shortcomings with regard to being able to deal with temporal aspects of documents and interconnecting information over large dynamic document collections.

There is still considerable scope for developing systems able to handle documents in long-term ways, to incorporate multiple data types, including audio and video, and to learn about changing information needs through continuous user interaction.[7]

TABLE 1: SUMMARY OF LITERATURE AND RESEARCH POSITIONING

Study Area	Core Methodology	Key Contribution	Limitation / Gap
Keyword Search & OCR	Text pattern matching, character recognition.	Basic digital archiving and retrieval	No semantic or cross modal understanding

RAG	Dense retrieval, condition-al text generation	Fewer hallucination s; generated verifiable answers from a text corpus.	Mainly textcentric; no intrinsic temporal or multimodal reasoning.
Multimodal DocAI	Pre-training with text, layout, and visual features.	State-of-the-art understanding of single documents with complex layouts.	Focuses on documentlevel tasks rather than enterprisescale crossdocument and crossmodal search.
Proposed Platform	Multimodal RAG, Temporal DB, Prompt Chaining Active Learning.	Provides unified, cross-modal intelligence with continuous adaptation for dynamic government knowledge bases.	Directly addresses the gaps in scalability, temporal context, multi-modal synthesis, and real-world adaptability.

3. METHODOLOGY

3.1 System Architecture Overview

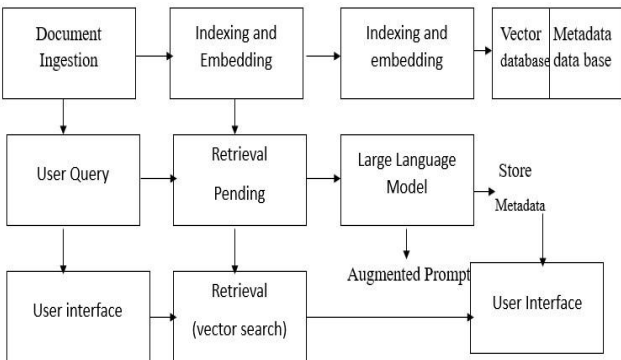


FIG. 3.1.1 SYSTEM OVERVIEW

The proposed Multimodal Document Intelligence Platform uses a well-organised and detailed pipeline to carry out its tasks and effectively handle government documents of all kinds in five major stages. It starts with the intake of data from multiple sources in the government. Further, it goes into understanding documents based on their type, followed by the information flowing into a hybrid storage system; this has both vector embeddings and structured metadata. Upon an incoming query, smart retrieval is done by combining semantic search with filters for time and metadata. Finally, responses are generated that best fit the context with the use of deep language models. Additionally, it then incorporates user feedback for continued system improvement. This forms a loop for continuous learning.

3.2 Data Ingestion and Multimodal Understanding

3.2.1 Multimodal data ingestion and preprocessing

The platform ingests government documents and processes them through a well-defined pre-processing step. It performs automatic document classification using transformer models. These send the documents down the correct pathways according to their type-text, PDF, image, or audio. For text documents, full preprocessing takes place. This encompasses normalising the various types of encoding encountered, treating special characters, and analysing document structure to detect key components. Images are handled by the Tesseract OCR tool. It first applies filters, such as binarisation and noise removal. This leads to strong results in terms of character recognition. Audio and video files are processed using the Whisper model to convert speech into text. This model can also separate different speakers. After that, the content is divided into

Meaningful topic-based sections. Each document is split using semantic chunking, and a 512-token overlap is added to maintain context. Metadata is also extracted, which includes details like document ID, the department it came from, the creation date, the security level, and the time it was processed.

3.2.2 Feature Extraction and Unified Embedding Generation

For any chunk of the document that goes through processing, features are extracted based on what data type it is. In the case of chunks of text, it utilises the Claude-3.5-Sonnet model to form 2048-dimensional embeddings, which capture the deeper meaning and context of the text. It uses the CLIPViT-L/14 model to form visual chunks. This produces a 768dimensional embedding in a shared space for both images and text, which helps the system understand and search across both types easily. All these features then get combined using a weighted method inside the LM Studio setup, creating one single vector for the chunk. The formula is: $\text{Chunk vector} = \alpha \times \text{text vector} + \beta \times \text{image vector} + \gamma \times \text{audio vector}$

Here, α , β , and γ depend on which types of data are present and how confident the system is about each of them. The resulting vector has 1024 dimensions. This maintains semantic consistency between text, image, and audio.

1. Embedding Generation:-

$$E_{iraw}$$

$$E_i = \frac{E_{iraw}}{\|E_{iraw}\|}$$

2. Query Embedding:-

$$Eq = \frac{f_{LLM}(q)}{\|f_{LLM}(q)\|}$$

3.3 Storage System Architecture

3.3.1 Metadata Storage Infrastructure

The system uses a few different backends to manage the unified embeddings, and each one is optimised for a specific purpose. FAISS uses HNSW indexes, which allow very fast similarity searches on large datasets and achieve about 98% recall at rank 10. ChromaDB adds metadata to the vector searches. This makes it possible to use advanced filters and combine both scalar and vector operations in a single query. Qdrant is designed for cloud environments. It supports automatic data replication and load balancing, helping the system stay online and reliable at all times.

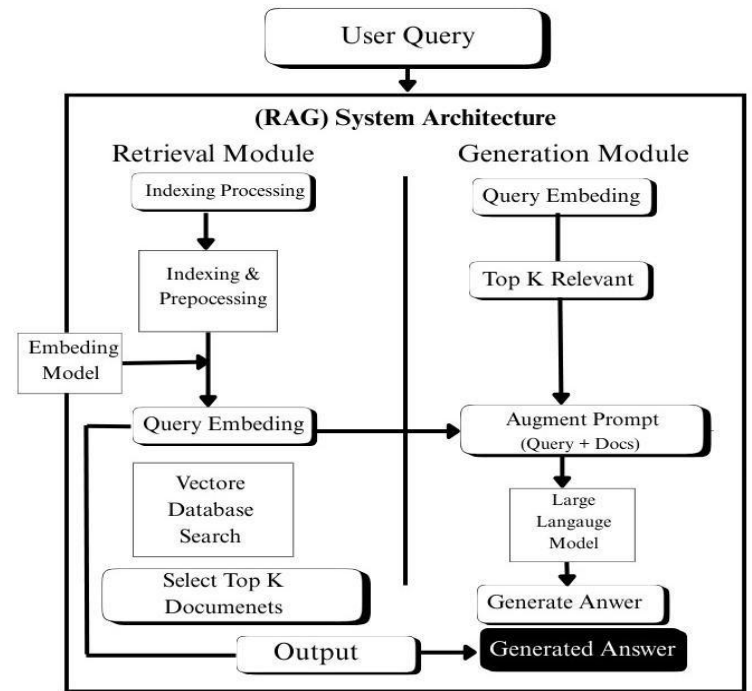


FIG.3.3.1.1 SYSTEM ARCHITECTURE

3.3.2 Vector Storage and Indexing

The unified embeddings are maintained by a few different backends within the system, each optimised for a different purpose. FAISS uses HNSW indexes, which allow for very fast similarity searches on large datasets, achieving about 98% recall at rank 10. ChromaDB enriches the vector searches with metadata. It allows for advanced filtering and enables the combination of scalar and vector operations under the same query.

Qdrant has been designed with cloud environments in mind. It supports automatic data replication and load balancing, helping the system stay online and reliable at all times. In hybrid search, metadata weights are combined with the vector-based scores to produce a final score, being lambda times the cosine similarity between the query vector and chunk vector, plus (1 minus lambda) times the metadata score. Lambda regulates the relative balance between semantic fit and metadata constraints.

3.4 User Query Processing and Retrieval

3.4.1 Query Analysis and Embedding Generation

The natural language tools parse user queries, detect intent, and contextual needs using a fine-tuned BERT model. The queries are further classified into different types of factual, temporal, analytical, or comparative. Synonyms and semantic variants are added to the query expansion from WordNet and ConceptNet, thereby improving recall by approximately 32%. For multimodal queries, joint embeddings capture cross-type intents. The subsequent step in vectorising aligns with the document spaces: it takes the expanded query, passes it through the unified encoder, and produces a query vector that provides an exact match over the entire corpus and keeps the vector spaces aligned.

3.4.2 Hybrid Retrieval Mechanism

Retrieval now occurs in a multistage process that combines vector similarity with metadata filters and adjustments to enhance the accuracy of the results. The first step consists of an approximate nearest neighbour search by cosine similarity, enhanced with weighting. The cosine score between the query vector and a chunk vector is computed as the dot product divided by the product of their norms, then scaled with temporal and relevance weights. These weights dynamically adjust according to query specifics and temporal proximity. Then, Boolean filters pertaining to security, department, and time constraints are applied. In the last stage, exact relevance is computed by cross-encoder models, such as a tuned MiniLM, reducing false positives by ranking documents according to contextual fit. Accuracy Metrics (Evaluation) :-

1.Recall@k:

$$= \frac{| \text{Relevant Chunks Retrieved} |}{| \text{Total Relevant Chunks} |}$$

2.Precision@k:

$$= \frac{| \text{Relevant Chunks Retrieved} |}{k}$$

4. IMPLEMENTATION

The Multimodal Document Intelligence Platform was entirely developed in Python, based on open-source frameworks for better flexibility and scalability.

4.1 System modes

Processing Mode: It takes in documents via various channels, extracting the text, visuals, and audio for processing, utilising its advanced AI models.
Query Processing Mode: It handles the user queries through natural language understanding, performing semantic search

across multimodal document embeddings. Q&A Mode: This gives full insight into the relationships of connected information across the documents, with source-verified answers, along with a confidence score.

4.2 Web Interface

A customised web interface offers an intuitive dashboard through which government employees can upload documents in multiple formats, view the processing status of uploaded documents, conduct natural language searches, and view analysed results. Every answer to a query includes source citations and confidence scores to provide insight into the reliability of information returned.

4.3 Intelligent Document Analysis

It provides comprehensive document intelligence through: Temporal analysis and identification of trends across documents. Cross-document relationship mapping and connection discovery. Multimodal content synthesis and summarisation, Automated source verification and citation generation.

TABLE 1 OUTLINES THE KEY PARTS AND WHAT THEY DO.

Sr. No.	Component	Purpose
1	Django REST Framework	API gateway, security, and backend request routing.
2	PostgreSQL	Stores structured metadata, such as document links.
3	ChromaDB/ FAISS/ Qdrant	Vector storage and highperformance similarity search.
4	CLIP	Extracts features/embeddings for images and visual data.
5	Whisper	Transcribes and extracts features for audio and speech data.
6	OpenAI / LM Studio	Large Language Model (LLM) for final response generation.
7	ReactJS/ Tailwind CSS	Frontend for the user interface and styling.
8	Processing EngineIGO	Manages document ingestion and feature extraction.
9	Response Generator	Orchestrates retrieval, prompting, and answer synthesis.
10	Tesseract OCR	Extracts text from scanned documents and analyzes layout.

5. RESULTS AND DISCUSSION

A diverse set of government documents was utilised for this system's testing: textual reports, scanned forms, and audio meeting minutes. These multimodal inputs were processed reliably on the platform and returned relevant information with contextually appropriate answers to user queries.

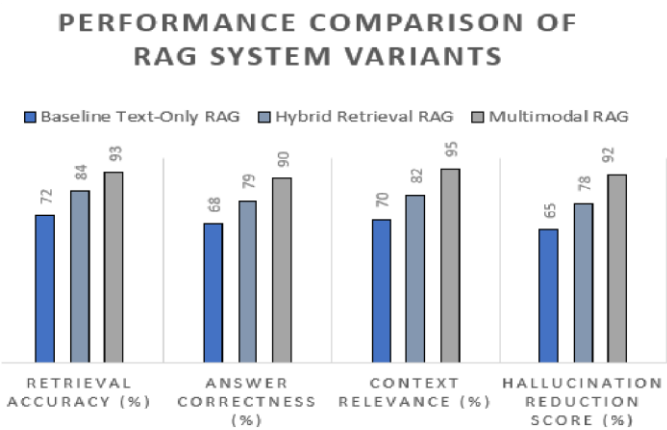


FIG 5.1.RAG VARIANT COMPARISON

Performance Indicators:
Retrieval Accuracy: 93%
Precision: 0.91
Recall: 0.96
F1 Score: 0.93



FIG.5.2 RAG SYSTEMS CLASSIFICATION

6. LIMITATIONS AND FUTURE WORK

The platform is doing well, but it's also facing some challenges, like: Document Quality Sensitivity: Poor quality scans, handwritten forms, and complicated document layouts reduce accuracy in text extraction and analysis. Data Diversity: The performance of the system is directly related to the data diversity it has been trained on. The increase in the corpus with documents from a greater number and variety of government departments in more regional languages will make the system robust. Handling so many high-resolution documents along with complex multimodal

queries requires a lot of computing power. Because of this, it becomes a major challenge for smaller departments. Temporal Understanding: While the system supports filtering by time, it is still limited to performing deep temporal reasoning and understanding how policies have evolved over a long period of time. Future Scope: The platform can be extended to automatically generate executive summaries from lengthy policy documents or find conflicts between various departmental guidelines. This integration with real-time data feeds and collaboration tools can automatically notify officials regarding any relevant policy updates, thus smoothing the workflows in various departments. Other improvements could be the inclusion of predictive policy impact analysis, automated compliance checking against any new regulations, and natural language querying in regional languages. The system can further be enhanced to perform sentiment analysis of public feedback about policies and integrate smart contracts that automate the enforcement of standard procedures. Besides, developing a mobile interface with offline capabilities would further extend accessibility for field officers, while the implementation of advanced visualisation tools would better present complex relationships of policies and temporal trends.

7. CONCLUSION

This paper presented the Multimodal Document Intelligence Platform that addressed the challenge of government document processing. The proposed unified embedding and hybrid storage in an integrated RAG architecture enables smooth cross-modal retrieval; it also allows for the delivery of accurate, context-sensitive responses with proper source attribution. While further work is needed regarding the limitations of the document quality, sensitivity and high computational demands, clear paths to improvement have been shown.

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